PHASE 1: STAGE 1 PREDICTIVE CORE (OFFLINE TRAINING)

Dataset Partition $\mathcal{D}_{\text{train}}$ (80%)

PaySim / IEEE-CIS / ULB:
Standardized Stratified Split (80/10/10)

Stage 1 XGBoost Predictive Core

Cost-Weighted Tree Ensemble $f_{\text{core}}(\mathbf{x})$
Deterministic Objective Optimization

Deploy f_{core}

PHASE 2: STAGE 2 VALIDATION & AMOUNT-STRATIFIED CALIBRATION

Validation Split $\mathcal{D}_{\text{val}}$ (10%)

$\hat{p}_{\text{val}} = \sigma(f_{\text{core}}(\mathbf{x}))$ via Held-Out Split

$k=5$ Stratified OOF Cross-Fitting

Extract Candidate Alerts ($\hat{p} \geq \tau_1$)
Local TreeSHAP $\phi(\mathbf{x})$ + Train RF Meta

OOF Preds q

Amount-Stratified Guard Calibration

1. Co-Optimized Operating Point τ_1:
PaySim: $\tau_1 = 0.1684$ | IEEE-CIS: $\tau_1 = 0.2800$

2. Stratify Alerts into K Amount Quantiles
3. Calibrate Strict Tier Thresholds:
$\tau_k = \max_{i \in \text{Fraud}_k} q_{\text{OOF},i} + \varepsilon$
4. Provable Zero False Negatives: $\Delta\text{FN} = 0$

PHASE 3: STAGE 3 PRODUCTION STREAMING INFERENCE WITH SAFETY GUARD

Incoming Transaction

$\mathbf{x} \in \mathbb{R}^d$ (Streaming)
Sub-millisecond Preproc

Stage 1 Fast-Path

Evaluate $f_{\text{core}}(\mathbf{x}) \to \hat{p}$
Latency: ~ 0.05 ms

$\hat{p} \geq \tau_1$

$\hat{p} < \tau_1$

Auto-Clear Legitimate ($\hat{p} < \tau_1$)

Direct Approval (> 99.7% of traffic)

Stage 3 TreeSHAP

Explain Alert ($\hat{p} \geq \tau_1$)
Exact Shapley $\phi(\mathbf{x})$

Joint Rep

Meta-Classifier Inference

Joint Space $[\mathbf{x}, \phi(\mathbf{x})]$
Output Risk Score q

Tier Guard Gate

Check $q \geq \tau_k(\text{Amt})$

$q \geq \tau_k$

Suppressed Alarm

Auto-Cleared FP
$\Delta\text{FN} = 0$ Guaranteed

$q < \tau_k$

Human Triage

High-Risk Fraud Alert
Prioritized Queue